

## Hormonal Control of Female Reproductive Cycles

Whereas sperm are produced continuously in human males, there are two closely linked reproductive cycles in human females. Both are controlled by cyclic patterns of endocrine signaling.

Cyclic events in the ovaries define the **ovarian cycle**: Once per cycle a follicle matures and an oocyte is released. Changes in the uterus define the **uterine cycle**, which in humans and some other primates is a menstrual cycle. In each **menstrual cycle**, the endometrium (lining of the uterus) thickens and develops a rich blood supply before being shed through the cervix and vagina if pregnancy does not occur. By linking the ovarian and uterine cycles, hormone activity synchronizes ovulation with the establishment of a uterine lining that can support embryo implantation and development.

If an oocyte is not fertilized and pregnancy does not occur, the uterine lining is sloughed off, and another pair of ovarian and uterine cycles begins. The cyclic shedding of the blood-rich endometrium from the uterus, a process that occurs in a flow through the cervix and vagina, is called **menstruation**. Menstrual (and ovarian) cycles average 28 days but can range from about 20 to 40 days. **Figure 46.14** uses the 28-day average to highlight the major events of these cycles for one ovarian and uterine cycle, illustrating the close coordination across different tissues of the reproductive system.

### The Ovarian Cycle

In human females, as in males, the hypothalamus has a central role in regulating reproduction. The ovarian cycle begins when the hypothalamus releases GnRH, which stimulates the anterior pituitary to secrete small amounts of FSH and LH.

▼ **Figure 46.14** The reproductive cycles of the human female. This figure shows how the ovarian cycle and the uterine (menstrual) cycle are regulated by changing hormone levels in the blood.

